

Smoke Test — King St E Section

Automated Pavement Inspection Report

Inspector: PaveScan AI Smoke Test | Date: May 06, 2026 at 11:31 PM | Images Analyzed: 3

■ IMMEDIATE ACTION REQUIRED

3 critical defect(s) detected that pose direct danger to road users. Emergency repair or barricading required within 24-48 hours.

Severities are AI suggestions — no inspector review applied.

Pavement Condition Index (PCI)

72

Satisfactory

Based on ASTM D6433 type-weighted deduct values with CDV correction

Very Poor (0-39)

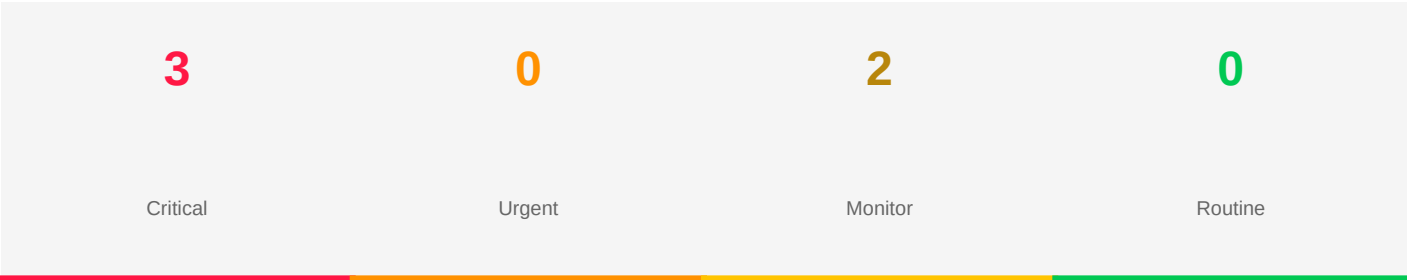
Poor (40-54)

Fair (55-69)

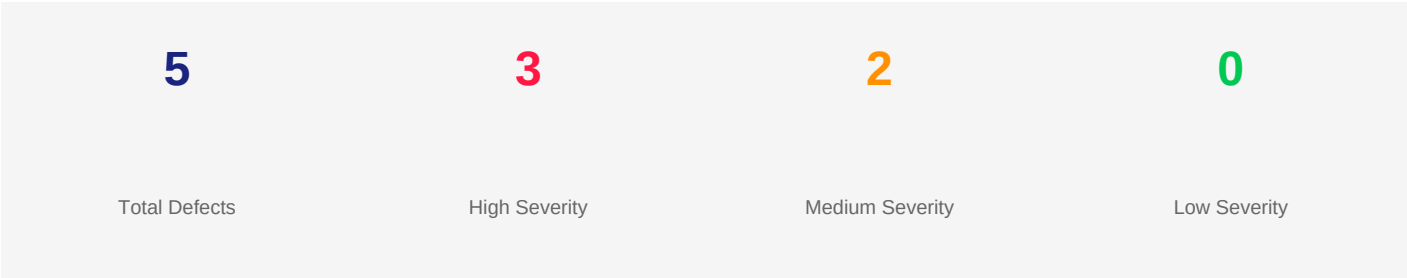
Satisfactory (70-84)

Good (85-100)

Safety Priority Summary



Executive Summary

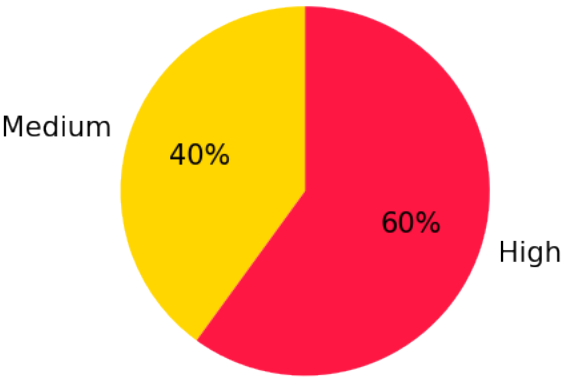


Average Confidence: 45.3%

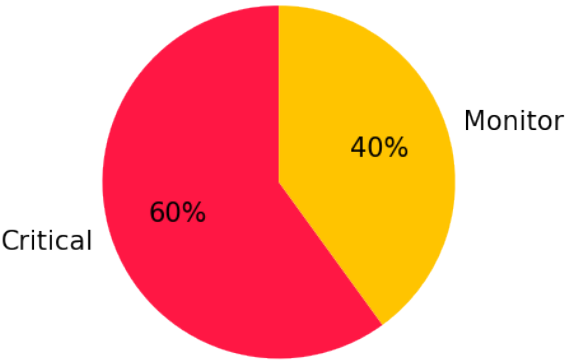
Images Analyzed: 3

Distribution Analysis

Severity Distribution



Safety Priority Distribution



Recommended Maintenance Plan

CRITICAL — Immediate — within 24-48 hours

Action: Emergency repair or barricade required

Defects (3): 3x crack

MONITOR — Planned — within 1-3 months

Action: Include in next maintenance cycle
Defects (2): 2x crack

■ **Cost Escalation Warning:** Deferred maintenance on critical and urgent defects typically increases repair costs by 4-8x. A pothole that costs \$50-150 to patch today may require \$2,000-5,000 in structural rehabilitation if left untreated through one freeze-thaw season. Early intervention is the most cost-effective strategy.

Defect Inventory

#	File	Defect Type	Priority	Confidence	Severity	Width (px)	Area (px)	Latitude	Longitude	Deterioration Risk
1	1686.rf.809fb1c5a1b607e5cf7dd718.jpg	Crack	CRITICAL	71%	HIGH (0.86)	663	53,951	43.660700	-79.395300	Unclassified crack → monitor for width progression and branching
2	1616.rf.c868709881a671796535255a.jpg	Crack	CRITICAL	53%	HIGH (0.78)	112	40,562	43.660500	-79.395500	Unclassified crack → monitor for width progression and branching
3	1675.rf.e3aa3f6c12d0247ef0jacc25.jpg	Crack	CRITICAL	25%	HIGH (0.73)	72	24,004	43.660600	-79.395400	Unclassified crack → monitor for width progression and branching
4	1675.rf.e3aa3f6c12d0247ef0jacc24.jpg	Crack	MONITOR	24%	MEDIUM (0.45)	45	8,745	43.660600	-79.395400	Unclassified crack → monitor for width progression and branching
5	1616.rf.c868709881a671796535255a.jpg	Crack	MONITOR	53%	MEDIUM (0.44)	44	7,966	43.660500	-79.395500	Unclassified crack → monitor for width progression and branching

Inspector Notes

Methodology

This report was generated using PaveScan AI, an automated pavement inspection system that combines deep learning-based defect detection with engineering assessment standards.

Detection: YOLO-based object detection and instance segmentation models trained on pavement distress imagery. Ensemble inference with IoU-based deduplication is used for improved coverage. Optional SAHI tiled inference detects small defects in high-resolution drone imagery.

Severity Assessment: Multi-factor scoring informed by ASTM D6433 Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys. Factors include defect area ratio, estimated crack width (from segmentation masks), defect type risk weight, detection confidence, and defect density. Structural failure types (potholes, alligator cracking) receive a minimum severity floor regardless of size.

Safety Priority: Four-tier classification (Critical, Urgent, Monitor, Routine) based on defect type, severity score, and systemic failure indicators. Prioritization follows a safety-first approach: all potholes are classified as critical regardless of size due to their direct hazard to road users.

PCI Scoring: Type-weighted deduct values with Corrected Deduct Value (CDV) diminishing returns, following the ASTM D6433 framework. Deducts are calibrated by defect type and severity level.

Note: This automated assessment supplements but does not replace professional engineering judgment. Critical findings should be verified by a qualified pavement engineer before major repair decisions.